"On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of those principles will lead to a zero grade and is subject to harsh discipline issues."

Ciağla Karacrslan

1597197

CamScanner ile tarandı

3)

a) $H^2 = I$

$H^T = H$

$H^2 = (I - 2vv^T)^2$

$= I^2 - 4vv^T + 4vvv^Tvv^T$

$v^Tv = 1$

$H^2 = I$ //

$v^Tv = 1$

b) $H^T = H$

$H^TH = HH$

$(I - 2vv^T)(I - 2vv^T)$

$= I - 4vv^T + 4v(v^Tv)v^T$

$I - 4vv^T + 4vv^T$

$= I \rightarrow H^TH = I$

$(H^T)^{-1} = H \rightarrow H^T = H^{-1}$

$H = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0$

$q_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |q| = \sqrt{2} \quad q_2 = \sqrt{2}$

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ → orthonormal matrix.

c) $Hv = (I - 2vv^T)v$

$= v - 2vv^Tv$

$= v - 2v$

$= -v$

$\lambda = -1$

d) For any vektor v orthogonal to v, we have

$Hv = (I - 2vv^T)v$

$= v - 2vv^Tv = v$ so: $\lambda = 1$ //

The orthogonal complement the space spanned by v has dimensions $(n-1)$ so the basis of $n-1$ ortonormal eigen vector. If we add eigenvector v we find, H is diagonalizable.

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1) $\alpha \to x$
$\gamma \to y$
a) $\beta \to z$

$$\begin{bmatrix} 1 & x & -y & x \\ 3x & 5 & -9y & 1 \\ 2 & z & 3 & x^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \quad \frac{1}{x} R_1 \to R_1$$

$$\begin{bmatrix} 1 & 1 & -y & 1 \\ 3x & 5 & -9y & 1 \\ 2 & z & 3 & x^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

b) $L \cdot U = A$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$\alpha = \frac{1}{3}$

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4)

a) $$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

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